

THE WORLD OF 2D SHAPES: VISUAL GUIDE & PROPERTIES

➤ **FLAT FIGURES: LENGTH & WIDTH ONLY** ➤

2D Shapes: Flat, two-dimensional geometric figures that exist only on a plane. They have length and width but no depth or thickness. Their properties include sides, vertices (corners), and angles.

1. POLYGONS (Straight Sides)

TRIANGLES (3 Sides, 3 Vertices, Sum of Angles = 180°)

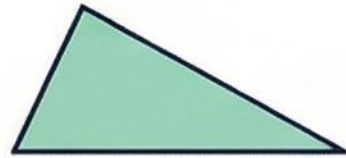
EQUILATERAL
(All sides & angles equal)



ISOSCELES
(Two equal sides & angles)



SCALENE
(No equal sides or angles)



QUADRILATERALS (4 Sides, 4 Vertices, Sum of Angles = 360°)

PARALLELOGRAMS
Two pairs of parallel sides



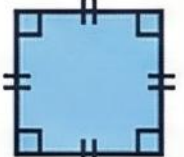
TRAPEZOID
(UK: Trapezium)
One pair of parallel sides



RECTANGLE
Four right angles, opposite sides equal



SQUARE
Four equal sides & right angles



RHOMBUS
Four equal sides, opposite angles equal



KITE
Two pairs of adjacent, equal-length sides

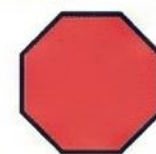


2. MORE POLYGONS (5+ Sides)

PENTAGON
(5 Sides, 5 Vertices)



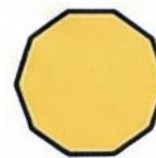
OCTAGON
(8 Sides, 8 Vertices)



HEXAGON
(6 Sides, 6 Vertices)



DECAGON
(10 Sides, 10 Vertices)



HEPTAGON
(7 Sides, 7 Vertices)



OCTAGON
(8 Sides, 8 Vertices)



DECAGON
(10 Sides, 10 Vertices)



3. NON-POLYGONS (Curved Lines)

CIRCLE
(One continuous curved line, all points equidistant from center)



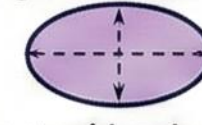
Infinite lines of symmetry.

OVAL / ELLIPSE
(Sides & corners)



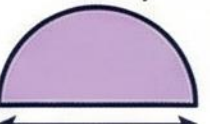
Two axes of symmetry.

OVAL / ELLIPSE
(Elongated curved shape, no straight sides or corners)



Erronate sides shape and refivment or corners.
Two axes of symmetry.

SEMICIRCLE
(Half a circle, one straight side, one curved)

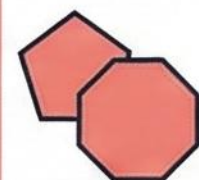


Diameter

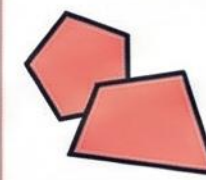
CLASSIFICATION & PROPERTIES

REGULAR vs. IRREGULAR

All sides and angles equal.



Sides and angles not all equal.



CONVEX

All interior angles $< 180^\circ$. No sides point inward.



vs.

CONCAVE

At least one interior angle $> 180^\circ$. Has a 'cave' or points inward.



THE WORLD OF 3D SHAPES: VISUAL GUIDE & PROPERTIES

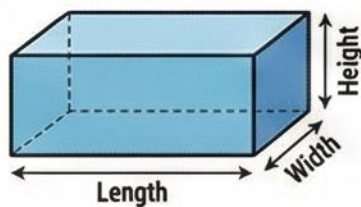
THREE-DIMENSIONAL FIGURES: LENGTH, WIDTH, & HEIGHT/DEPTH

3D Shapes: Solid geometric figures that exist in three dimensions. They have length, width, and height (or depth). Their main properties are **faces** (flat surfaces), **edges** (where faces meet), and **vertices** (corners where edges meet).

1 POLYHEDRONS (Flat Faces, Straight Edges)

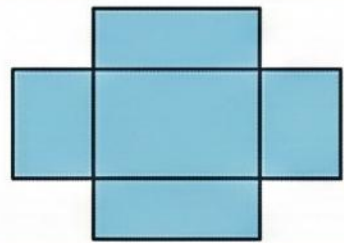
PRISMS (Two identical, parallel bases & rectangular faces)

RECTANGULAR PRISM (CUBOID)



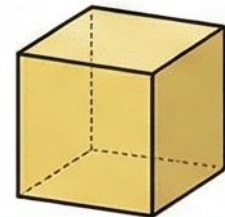
Properties

Faces: 6
Edges: 12
Vertices: 8
Net included.



CUBE

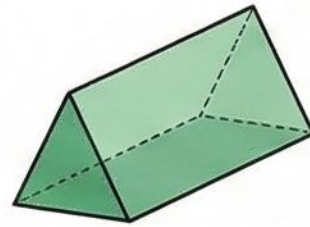
(All faces are squares)



Properties

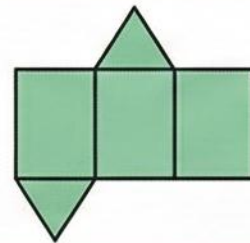
Faces: 6
Edges: 12
Vertices: 8
Special type of rectangular prism.

TRIANGULAR PRISM



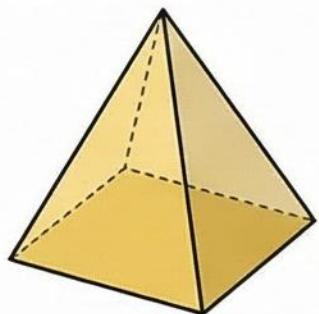
Properties

Faces: 5
Edges: 9
Vertices: 6
Net included.

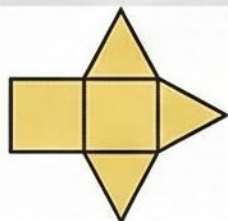


PYRAMIDS (One base & triangular faces meeting at a point)

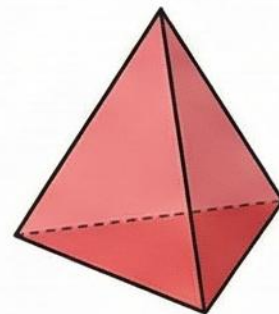
SQUARE PYRAMID



Faces: 5
Edges: 8
Vertices: 5. Net included.



TRIANGULAR PYRAMID (TETRAHEDRON)

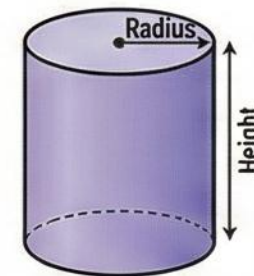


Faces: 4
Edges: 6
Vertices: 4
All faces are triangles.

2 NON-POLYHEDRONS (Curved Surfaces)

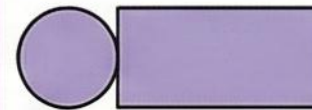
CYLINDER

(Two parallel circular bases & one curved surface)



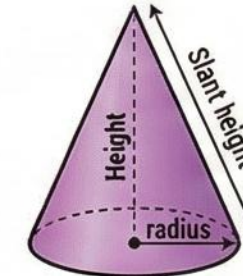
Properties

Faces: 2 flat, 1 curved
Edges: 2 curved
Vertices: 0
Net included.



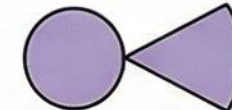
CONE

(One circular base, one curved surface, & an apex)



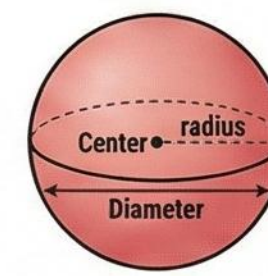
Properties

Faces: 1 flat, 1 curved
Edges: 1 curved
Vertices: 1 (Apex).
Net included.



SPHERE

(A perfectly round geometrical object in 3D space)



Properties

Faces: 0 flat, 1 curved
Edges: 0
Vertices: 0
All points on surface equidistant from center.

TORUS

(A surface of revolution generated by revolving a circle in three-dimensional space about an axis coplanar with the circle)



Properties

Faces: 1 curved surface
Edges: 0
Vertices: 0
Has a hole.

3 KEY PROPERTIES SUMMARY



FACES (F)

A flat or curved surface of a 3D shape.



EDGES (E)

The line segment where two faces meet.



VERTICES (V)

The point where three or more edges meet.

EULER'S FORMULA (For Convex Polyhedrons): $F + V - E = 2$

